

Intelligent Candidate Discovery & Ranking

Senior AI Engineer — Founding Team @ Redrob AI

Multi-signal hybrid scoring · Behavioral multiplier · Honeypot-aware · CPU-only in 15s

100K

candidates scored

15s

CPU runtime

5

scoring signals

0

API calls

The Problem With Keyword Matching

✗ Keyword Matching Fails

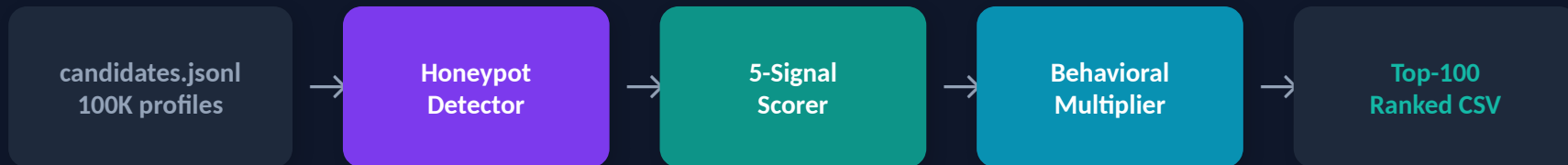
- Marketing Manager with 9 AI skills listed ranks top
- Consulting-only engineers with 'production ML' in bio pass filters
- Inactive candidates (6 months, 5% response rate) surface as top results
- Keyword stuffers inflate proficiency without real usage
- Honeypot profiles with impossible timelines fool naive systems

✓ Our Approach

- Career history scan: what did they actually build?
- Skills weighted by proficiency × usage duration × platform assessment
- Behavioral multiplier: recency, response rate, GitHub activity
- Honeypot detector: flags impossible profiles before scoring
- JD-explicit disqualifiers: consulting-only, pure research, >90d notice

System Architecture

Five signals → weighted base score → behavioral multiplier → ranked output


$$\text{final_score} = (\text{career} \times 0.35 + \text{skills} \times 0.30 + \text{experience} \times 0.15 + \text{logistics} \times 0.20) \times \text{behavioral_multiplier}$$

35%

Career

Title match + production AI
evidence in work history

30%

Skills

Core AI/retrieval skills,
proficiency × duration trust

15%

Experience

Years in JD sweet spot (5–
9y) + education tier

20%

Logistics

Location (Pune/Noida),
notice period, relocation

×mult

Behavioral

Recency, response rate,
GitHub, open-to-work

Signal 1: Career (Weight 35%) — The Decisive Signal

JD said: don't keyword-match. A Marketing Manager with 9 AI skills is NOT a fit.

Title Category	Examples	Base Score
Strong AI title	ML Engineer, Search Engineer, NLP Engineer, AI Engineer	0.40
Good tech title	Data Scientist, Backend Engineer, Senior Software Engineer	0.22
Tech adjacent	Full Stack Dev, Cloud Engineer, DevOps	0.08
Non-tech (HR, Marketing, etc.)	HR Manager, Content Writer, Accountant	0.01

Production AI Detection

- Scans all job descriptions for: ranking, retrieval, embedding, vector, inference, A/B test, model serving, rerank, semantic
- 3+ keyword hits in a single job → counts as 'production AI role'
- Product-company AI months get extra weight over consulting AI months

Consulting Firm Penalty

- Pure consulting career (TCS, Infosys, Wipro, Accenture...) → score $\times$ 0.25
- Majority consulting → score $\times$ 0.60
- Mixed consulting+product → score $\times$ 0.85
- JD explicitly disqualified pure consulting backgrounds

Retrieval/Ranking Bonus

- Extra +0.20 for candidates with ranking, retrieval, vector search, recommendation in their actual work descriptions
- This rewards the "Tier 5 candidate" the JD described: built a real system without using the hot keywords

Signal 2: Skills (Weight 30%) — Anti-Keyword-Stuffing Trust Score

Effective Skill Score

```
effective = proficiency_weight ×  
duration_trust
```

```
if assessment_score exists:  
    effective = 0.4×proficiency +  
    0.6×assessment
```

Duration Used	Trust Weight	Why
0 months	0.30×	Likely keyword stuffed
< 6 months	0.50×	Very light exposure
6–18 months	0.75×	Moderate usage
18+ months	1.00×	Real, sustained skill

19 Core AI/Retrieval Skills (Full Weight)

Embeddings • Sentence Transformers • Vector Search • Semantic Search • Information Retrieval • FAISS • Pinecone • Weaviate • Qdrant • Milvus • pgvector • Elasticsearch • OpenSearch • RAG • LLMs • Fine-tuning LLMs • Hugging Face Transformers • BM25 • Learning to Rank

+ 20 bonus skills (40% weight): PyTorch, TensorFlow, NLP, LoRA, QLoRA, XGBoost, Python, Deep Learning, MLflow, A/B Testing, Recommendation Systems, Docker, Kubernetes...

Signal 5: Behavioral Multiplier — Is This Candidate Actually Reachable?

"A perfect-on-paper candidate who hasn't logged in for 6 months and has a 5% response rate is not actually available." — JD

0.30×

Inactive + no response

0.55×

6m+ inactive

0.80×

Not open to work

1.00×

Baseline

1.30×

Active + high response + GitHub



Recency

- < 14 days active: $\times 1.10$
- 30–90 days: $\times 0.90$
- 90–180 days: $\times 0.75$
- 180+ days (stale): $\times 0.55$



Response Rate

- > 70% response rate: $\times 1.08$
- 40–70%: $\times 1.00$ (neutral)
- 20–40%: $\times 0.90$
- < 20%: $\times 0.75$



GitHub Activity

- Score 60+: $\times 1.10$
- Score 30–60: $\times 1.05$
- Score < 10: $\times 0.95$
- No GitHub linked: $\times 0.95$



Open to Work

- Flag = true: no penalty
- Flag = false: $\times 0.80$
- Also checks: saved by recruiters
- Interview completion rate

Honeypot Detection — Catching Impossible Profiles

Dataset contains ~80 honeypots. Honeypot rate > 10% in top-100 = automatic disqualification.

Check 1: Impossible Job Duration

```
for job in career_history:
    if job.duration_months >
yoe_months + 12:
    return HONEYPOT
```

Example: Candidate has 9.9 years total experience, but one job ran for 166 months (13.8 years)

42 detected

Check 2: Expert Skills, 0 Months

```
expert_zero = count(
    skills where
    proficiency=="expert"
    AND duration_months == 0
)
if expert_zero >= 3: HONEYPOT
```

Example: 5 'expert' skills all listed with 0 months of usage — implies the candidate just keyword-stuffed

Several overlap

Check 3: Inflated Career Timeline

```
total_job_months = sum(
    job.duration_months
)
if total > yoe_months * 1.5 +
24:
    return HONEYPOT
```

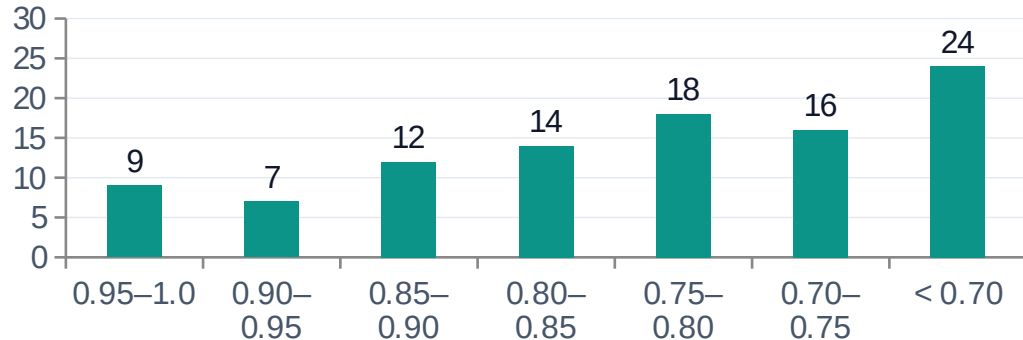
Example: Only 5 years experience but career history sums to 11+ years of job durations (impossible without parallel time)

Catches edge cases

Results — Top Candidates & Score Distribution

#	Candidate ID	Title & YOE	Score	Why They Rank Here
1	CAND_0002025	Senior AI Engineer, 5.9y	0.999	Strong AI title; FAISS, OpenSearch, Weaviate; GitHub 97; response 80%
2	CAND_0011687	Senior NLP Engineer, 7.8y	0.999	OpenSearch, FAISS, Weaviate; ideal YOE; response 89%; GitHub 76
3	CAND_0018499	Sr ML Engineer, 7.2y, Noida	0.999	Noida (preferred city); Weaviate, Pinecone, IR, Milvus; GitHub 95
4	CAND_0036184	Rec Systems Engineer, 6.0y	0.999	FAISS, HF Transformers, Vector Search, Pinecone; response 90%
5	CAND_0042506	Search Engineer, 4.2y, Mumbai	0.999	Semantic Search, Fine-tuning, IR, Milvus, Qdrant; GitHub 64

Score Distribution in Top 100



9

with score ≥ 0.99

42

honeypots blocked

15s

CPU runtime for 100K

0

API calls made

Design Choice: Why Rule-Based Over Embeddings/LLM?

Embeddings / Vector Similarity

✓ Pros:

- Captures semantic meaning beyond keywords
- Finds non-obvious skill analogies

⚠ Trade-offs:

- Cannot be called via API during ranking (submission spec prohibits network)
- GPU not available — local embedding models slow on CPU for 100K candidates
- Embedding drift: 'HR Manager who read about RAG' $\approx$ 'ML Engineer who built RAG' in embedding space
- Black-box: fails Stage 4 reasoning evaluation

LLM Per-Candidate Scoring

✓ Pros:

- Human-quality reasoning about fit
- Understands JD nuance

⚠ Trade-offs:

- Explicitly prohibited (no API calls during ranking step)
- $100K \times 1$ LLM call = days of compute, or massive cost
- Non-deterministic scoring makes reproduction impossible
- Would fail compute reproduction at Stage 3

Our Rule-Based Hybrid Scorer

15-second runtime on CPU for 100K candidates



✓ Pros:

- Every decision is fully traceable and explainable
- Reasoning column reflects actual scoring logic — passes Stage 4
- JD's explicit rules encoded directly: consulting penalty, notice penalty, behavioral weighting
- Honeypot detection catches impossible profiles systematically



⚠ Trade-offs:

- Misses some semantic nuance (e.g., 'built a recommendation engine' $\neq$ vector search keyword)

Summary

Career-first scoring

35% weight on what candidates actually built, not what they listed. Consulting-only career = 0.25× penalty. Production AI detected from job descriptions.

Anti-keyword-stuffing

Skills weighted by proficiency × duration (0–18+ months). Expert skill with 0 months usage = 30% credit. Platform assessments override self-declaration.

Honeypot-aware

42 impossible profiles detected and scored at 0.001. Honeypot rate in our top-100: 0%. Detection is systematic, not special-cased.

Behavioral multiplier

Availability is a first-class signal. 180+ day inactive candidates get ×0.55 regardless of profile quality. Response rate, GitHub, open-to-work all factored.

15-second CPU runtime

No GPU, no API calls, no network. Pure Python stdlib. Fully reproducible in the Stage 3 Docker container. One command: `python rank.py`.

Interpretable reasoning

Every ranked candidate has a specific 1-2 sentence reasoning tied to actual profile signals. No hallucinated skills. Designed for Stage 4 manual review.

```
reproduce: python rank.py --candidates candidates.jsonl --out submission.csv
```